

## Experiment 1

**Aim:** To perform and understand fundamental DBMS operations, including database creation, table management, data manipulation, modification, and retrieval using MySQL.

1. SHOW databases;

```
MySQL localhost:33060+ ssl SQL > SHOW DATABASES;
+-----+
| Database |
+-----+
| college  |
| company  |
| information_schema |
| mysql    |
| performance_schema |
| sakila   |
| sys      |
| world    |
+-----+
8 rows in set (0.0019 sec)
```

2. - CREATE DATABASE stud1test;  
- SHOW DATABASES;

```
MySQL localhost:33060+ ssl SQL > CREATE DATABASE stud1_test;
Query OK, 1 row affected (0.0129 sec)
MySQL localhost:33060+ ssl SQL > SHOW DATABASES;
+-----+
| Database |
+-----+
| college  |
| company  |
| information_schema |
| mysql    |
| performance_schema |
| sakila   |
| stud1_test |
| sys      |
| world    |
+-----+
9 rows in set (0.0016 sec)
```

3. - DROP DATABASE stud1\_test;  
- SHOW DATABASES;

```
MySQL localhost:33060+ ssl SQL > DROP DATABASE stud1_test;
Query OK, 0 rows affected (0.0178 sec)
MySQL localhost:33060+ ssl SQL > SHOW DATABASES;
+-----+
| Database |
+-----+
| college  |
| company  |
| information_schema |
| mysql    |
| performance_schema |
| sakila   |
| sys      |
| world    |
+-----+
8 rows in set (0.0020 sec)
```

4. - CREATE DATABASE aarya61;  
- SHOW DATABASES;

```
MySQL localhost:33060+ ssl SQL > CREATE DATABASE aarya61;
Query OK, 1 row affected (0.0095 sec)
MySQL localhost:33060+ ssl SQL > SHOW DATABASES;
+-----+
| Database |
+-----+
| aarya61  |
| college  |
| company  |
| information_schema |
| mysql     |
| performance_schema |
| sakila    |
| sys       |
| world     |
+-----+
9 rows in set (0.0018 sec)
```

5. - CREATE TABLE Persons ( PersonID int, LastName varchar(25),  
FirstName varchar(25), Address varchar(100), City varchar(25) );  
- SHOW TABLES;  
- DESCRIBE Persons;

```
MySQL localhost:33060+ ssl SQL > USE aarya61;
Default schema set to 'aarya61'.
Fetching global names, object names from 'aarya61' for auto-completion...
MySQL localhost:33060+ ssl aarya61 SQL > CREATE TABLE Persons (
->     PersonID int,
->     LastName varchar(25),
->     FirstName varchar(25),
->     Address varchar(100),
->     City varchar(25)
-> );
Query OK, 0 rows affected (0.0237 sec)
MySQL localhost:33060+ ssl aarya61 SQL >
MySQL localhost:33060+ ssl aarya61 SQL > SHOW TABLES;
+-----+
| Tables_in_aarya61 |
+-----+
| persons            |
+-----+
1 row in set (0.0015 sec)
MySQL localhost:33060+ ssl aarya61 SQL > DESCRIBE Persons;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| PersonID   | int           | YES  |     | NULL    |       |
| LastName    | varchar(25)   | YES  |     | NULL    |       |
| FirstName   | varchar(25)   | YES  |     | NULL    |       |
| Address     | varchar(100)  | YES  |     | NULL    |       |
| City        | varchar(25)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.0081 sec)
```

6. - DROP TABLE Persons;
- CREATE TABLE Customers ( CustomerID int, LastName varchar(25), FirstName varchar(25), Address varchar(100), City varchar(25), PRIMARY KEY (CustomerID) );
- DESCRIBE Customers;

```
MySQL localhost:33060+ ssl aarya61 SQL > DROP TABLE Persons;
Query OK, 0 rows affected (0.0119 sec)
MySQL localhost:33060+ ssl aarya61 SQL >
MySQL localhost:33060+ ssl aarya61 SQL > CREATE TABLE Customers (
-> CustomerID int,
-> LastName varchar(25),
-> FirstName varchar(25),
-> Address varchar(100),
-> City varchar(25),
-> PRIMARY KEY (CustomerID)
-> );
Query OK, 0 rows affected (0.0186 sec)
MySQL localhost:33060+ ssl aarya61 SQL >
MySQL localhost:33060+ ssl aarya61 SQL > DESCRIBE Customers;
```

| Field      | Type         | Null | Key | Default | Extra |
|------------|--------------|------|-----|---------|-------|
| CustomerID | int          | NO   | PRI | NULL    |       |
| LastName   | varchar(25)  | YES  |     | NULL    |       |
| FirstName  | varchar(25)  | YES  |     | NULL    |       |
| Address    | varchar(100) | YES  |     | NULL    |       |
| City       | varchar(25)  | YES  |     | NULL    |       |

5 rows in set (0.0035 sec)

7. - INSERT INTO Customers (CustomerID, LastName, FirstName, Address, City) VALUES (96365, 'Shah', 'Hiren', 'A67,Building10,abc lane', 'Mumbai');
- SELECT \* FROM Customers;

```
MySQL localhost:33060+ ssl aarya61 SQL > INSERT INTO Customers (CustomerID, LastName, FirstName, Address, City)
-> VALUES (96365, 'Shah', 'Hiren', 'A67,Building10,abc lane', 'Mumbai');
Query OK, 1 row affected (0.0077 sec)
MySQL localhost:33060+ ssl aarya61 SQL >
MySQL localhost:33060+ ssl aarya61 SQL > SELECT * FROM Customers;
```

| CustomerID | LastName | FirstName | Address                 | City   |
|------------|----------|-----------|-------------------------|--------|
| 96365      | Shah     | Hiren     | A67,Building10,abc lane | Mumbai |

1 row in set (0.0011 sec)

8. - INSERT INTO Customers VALUES (96300, 'Singh', 'Prakesh', 'B7,House10,abc lane', 'Mumbai'), (89790, 'Joshi', 'Ramesh', '413,Sadashiv Nagar', 'Pune');

- SELECT \* FROM Customers;

```
MySQL localhost:33060+ ssl aarya61 SQL > INSERT INTO Customers VALUES
-> (96300, 'Singh', 'Prakesh', 'B7,House10,abc lane', 'Mumbai'),
-> (89790, 'Joshi', 'Ramesh', '413,Sadashiv Nagar', 'Pune');

Query OK, 2 rows affected (0.0060 sec)

Records: 2 Duplicates: 0 Warnings: 0
MySQL localhost:33060+ ssl aarya61 SQL >
MySQL localhost:33060+ ssl aarya61 SQL > SELECT * FROM Customers;
+-----+-----+-----+-----+-----+
| CustomerID | LastName | FirstName | Address | City |
+-----+-----+-----+-----+-----+
| 89790 | Joshi | Ramesh | 413,Sadashiv Nagar | Pune |
| 96300 | Singh | Prakesh | B7,House10,abc lane | Mumbai |
| 96365 | Shah | Hiren | A67,Building10,abc lane | Mumbai |
+-----+-----+-----+-----+-----+
3 rows in set (0.0011 sec)
```

## 9. - INSERT INTO Customers VALUES

(96301, 'Patil', 'Aditya', '12,MG Road', 'Mumbai'),  
 (96302, 'Mehta', 'Riya', '45,FC Road', 'Pune'),  
 (96303, 'Desai', 'Om', '78,Station Road', 'Nashik'),  
 (96304, 'Kulkarni', 'Sneha', '21,JM Road', 'Pune'),  
 (96305, 'Sharma', 'Rahul', '56,Link Road', 'Mumbai'),  
 (96306, 'Joshi', 'Neha', '89,College Road', 'Nashik'),  
 (96307, 'Pawar', 'Amit', '34,Main Road', 'Thane');

- SELECT \* FROM Customers;

```
MySQL localhost:33060+ ssl aarya61 SQL > INSERT INTO Customers VALUES
-> (96301, 'Patil', 'Aditya', '12,MG Road', 'Mumbai'),
-> (96302, 'Mehta', 'Riya', '45,FC Road', 'Pune'),
-> (96303, 'Desai', 'Om', '78,Station Road', 'Nashik'),
-> (96304, 'Kulkarni', 'Sneha', '21,JM Road', 'Pune'),
-> (96305, 'Sharma', 'Rahul', '56,Link Road', 'Mumbai'),
-> (96306, 'Joshi', 'Neha', '89,College Road', 'Nashik'),
-> (96307, 'Pawar', 'Amit', '34,Main Road', 'Thane');

Query OK, 7 rows affected (0.0055 sec)

Records: 7 Duplicates: 0 Warnings: 0
MySQL localhost:33060+ ssl aarya61 SQL >
MySQL localhost:33060+ ssl aarya61 SQL > SELECT * FROM Customers;
+-----+-----+-----+-----+-----+
| CustomerID | LastName | FirstName | Address | City |
+-----+-----+-----+-----+-----+
| 89790 | Joshi | Ramesh | 413,Sadashiv Nagar | Pune |
| 96300 | Singh | Prakesh | B7,House10,abc lane | Mumbai |
| 96301 | Patil | Aditya | 12,MG Road | Mumbai |
| 96302 | Mehta | Riya | 45,FC Road | Pune |
| 96303 | Desai | Om | 78,Station Road | Nashik |
| 96304 | Kulkarni | Sneha | 21,JM Road | Pune |
| 96305 | Sharma | Rahul | 56,Link Road | Mumbai |
| 96306 | Joshi | Neha | 89,College Road | Nashik |
| 96307 | Pawar | Amit | 34,Main Road | Thane |
| 96365 | Shah | Hiren | A67,Building10,abc lane | Mumbai |
+-----+-----+-----+-----+-----+
10 rows in set (0.0016 sec)
```

10. - ALTER TABLE Customers ADD Email varchar(50);
- DESCRIBE Customers;
- SELECT \* FROM Customers;

```
MySQL localhost:33060+ ssl aarya61 SQL> ALTER TABLE Customers ADD Email varchar(50);
Query OK, 0 rows affected (0.0233 sec)

Records: 0 Duplicates: 0 Warnings: 0
MySQL localhost:33060+ ssl aarya61 SQL>
MySQL localhost:33060+ ssl aarya61 SQL> DESCRIBE Customers;
```

| Field      | Type         | Null | Key | Default | Extra |
|------------|--------------|------|-----|---------|-------|
| CustomerID | int          | NO   | PRI | NULL    |       |
| LastName   | varchar(25)  | YES  |     | NULL    |       |
| FirstName  | varchar(25)  | YES  |     | NULL    |       |
| Address    | varchar(100) | YES  |     | NULL    |       |
| City       | varchar(25)  | YES  |     | NULL    |       |
| Email      | varchar(50)  | YES  |     | NULL    |       |

6 rows in set (0.0033 sec)

```
MySQL localhost:33060+ ssl aarya61 SQL> SELECT * FROM Customers;
```

| CustomerID | LastName | FirstName | Address                 | City   | Email |
|------------|----------|-----------|-------------------------|--------|-------|
| 89790      | Joshi    | Ramesh    | 413,Sadashiv Nagar      | Pune   | NULL  |
| 96300      | Singh    | Prakesh   | 87,House10,abc lane     | Mumbai | NULL  |
| 96301      | Patil    | Aditya    | 12,MG Road              | Mumbai | NULL  |
| 96302      | Mehta    | Riya      | 45,FC Road              | Pune   | NULL  |
| 96303      | Desai    | Om        | 78,Station Road         | Nashik | NULL  |
| 96304      | Kulkarni | Sneha     | 21,JM Road              | Pune   | NULL  |
| 96305      | Sharma   | Rahul     | 56,Link Road            | Mumbai | NULL  |
| 96306      | Joshi    | Neha      | 89,College Road         | Nashik | NULL  |
| 96307      | Pawar    | Amit      | 34,Main Road            | Thane  | NULL  |
| 96365      | Shah     | Hiren     | A67,Building10,abc lane | Mumbai | NULL  |

10 rows in set (0.0015 sec)

11. - ALTER TABLE Customers ADD Remark varchar(255);
- DESCRIBE Customers;
- SELECT \* FROM Customers;

```
MySQL localhost:33060+ ssl aarya61 SQL> ALTER TABLE Customers ADD Remark varchar(255);
Query OK, 0 rows affected (0.0127 sec)

Records: 0 Duplicates: 0 Warnings: 0
MySQL localhost:33060+ ssl aarya61 SQL>
MySQL localhost:33060+ ssl aarya61 SQL> DESCRIBE Customers;
```

| Field      | Type         | Null | Key | Default | Extra |
|------------|--------------|------|-----|---------|-------|
| CustomerID | int          | NO   | PRI | NULL    |       |
| LastName   | varchar(25)  | YES  |     | NULL    |       |
| FirstName  | varchar(25)  | YES  |     | NULL    |       |
| Address    | varchar(100) | YES  |     | NULL    |       |
| City       | varchar(25)  | YES  |     | NULL    |       |
| Email      | varchar(50)  | YES  |     | NULL    |       |
| Remark     | varchar(255) | YES  |     | NULL    |       |

7 rows in set (0.0013 sec)

```
MySQL localhost:33060+ ssl aarya61 SQL> SELECT * FROM Customers;
```

| CustomerID | LastName | FirstName | Address                 | City   | Email | Remark |
|------------|----------|-----------|-------------------------|--------|-------|--------|
| 89790      | Joshi    | Ramesh    | 413,Sadashiv Nagar      | Pune   | NULL  | NULL   |
| 96300      | Singh    | Prakesh   | 87,House10,abc lane     | Mumbai | NULL  | NULL   |
| 96301      | Patil    | Aditya    | 12,MG Road              | Mumbai | NULL  | NULL   |
| 96302      | Mehta    | Riya      | 45,FC Road              | Pune   | NULL  | NULL   |
| 96303      | Desai    | Om        | 78,Station Road         | Nashik | NULL  | NULL   |
| 96304      | Kulkarni | Sneha     | 21,JM Road              | Pune   | NULL  | NULL   |
| 96305      | Sharma   | Rahul     | 56,Link Road            | Mumbai | NULL  | NULL   |
| 96306      | Joshi    | Neha      | 89,College Road         | Nashik | NULL  | NULL   |
| 96307      | Pawar    | Amit      | 34,Main Road            | Thane  | NULL  | NULL   |
| 96365      | Shah     | Hiren     | A67,Building10,abc lane | Mumbai | NULL  | NULL   |

10 rows in set (0.0015 sec)

12. - UPDATE Customers
- SET Email =
- 'ramesh.joshi@abc.com', Remark
- = 'Regular Buyer'
- WHERE CustomerID = 89790;
- SELECT \* FROM Customers

WHERE CustomerID = 89790;

```
MySQL localhost:33060+ ssl aarya61 SQL > UPDATE Customers
-> SET Email = 'ramesh.joshi@abc.com',
-> Remark = 'Regular Buyer'
-> WHERE CustomerID = 89790;
Query OK, 1 row affected (0.0083 sec)

Rows matched: 1 Changed: 1 Warnings: 0
MySQL localhost:33060+ ssl aarya61 SQL > SELECT * FROM Customers
-> WHERE CustomerID = 89790;
```

| CustomerID | LastName | FirstName | Address            | City | Email                | Remark        |
|------------|----------|-----------|--------------------|------|----------------------|---------------|
| 89790      | Joshi    | Ramesh    | 413,Sadashiv Nagar | Pune | ramesh.joshi@abc.com | Regular Buyer |

1 row in set (0.0011 sec)

13. SELECT CustomerID, Address FROM Customers;

```
MySQL localhost:33060+ ssl aarya61 SQL > SELECT CustomerID, Address FROM Customers;
```

| CustomerID | Address                 |
|------------|-------------------------|
| 89790      | 413,Sadashiv Nagar      |
| 96300      | B7,House10,abc lane     |
| 96301      | 12,MG Road              |
| 96302      | 45,FC Road              |
| 96303      | 78,Station Road         |
| 96304      | 21,JM Road              |
| 96305      | 56,Link Road            |
| 96306      | 89,College Road         |
| 96307      | 34,Main Road            |
| 96365      | A67,Building10,abc lane |

10 rows in set (0.0017 sec)

14. SELECT CustomerID,  
Address FROM Customers  
WHERE CustomerID = 96300;

```
MySQL localhost:33060+ ssl aarya61 SQL > SELECT CustomerID, Address
-> FROM Customers
-> WHERE CustomerID = 96300;
```

| CustomerID | Address             |
|------------|---------------------|
| 96300      | B7,House10,abc lane |

1 row in set (0.0014 sec)

15. SELECT \* FROM  
Customers WHERE  
CustomerID = 96300;

```
MySQL localhost:33060+ ssl aarya61 SQL > SELECT * FROM Customers
-> WHERE CustomerID = 96300;
```

| CustomerID | LastName | FirstName | Address             | City   | Email | Remark |
|------------|----------|-----------|---------------------|--------|-------|--------|
| 96300      | Singh    | Prakesh   | B7,House10,abc lane | Mumbai | NULL  | NULL   |

1 row in set (0.0017 sec)

**Q1. What is a database? Why do we need a database management system such as MySQL?**

A database is an organized collection of data. A DBMS such as MySQL is used to store, manage, update, and retrieve data efficiently.

**Q2. What is the difference between CREATE DATABASE, SHOW DATABASES, and DROP DATABASE?**

| Command         | Purpose                                    |
|-----------------|--------------------------------------------|
| CREATE DATABASE | Creates a new database.                    |
| SHOW DATABASES  | Displays all available databases.          |
| DROP DATABASE   | Deletes an existing database and its data. |

**Q3. What is a table in a database? Explain the purpose of rows and columns in a table.**

A table is used to store data in an organized form.

- **Rows:** Represent individual records.
- **Columns:** Represent attributes or properties of the records.

For example, in the Customers table, each row represents one customer, while CustomerID, FirstName, and City are columns.

**Q4. What is a primary key? Why was CustomerID selected as the primary key in the Customers table?**

A primary key is a column that uniquely identifies each record in a table. CustomerID was selected as the primary key because every customer can have a unique ID. A primary key cannot contain duplicate or NULL values.

**Q5. What is the purpose of the VARCHAR and INT data types? Give one suitable example of each.**

- **INT:** Used to store whole numbers.  
**Example:** CustomerID INT
- **VARCHAR:** Used to store variable-length text.  
**Example:** FirstName VARCHAR(25)

**Q6. What is the difference between INSERT, SELECT, UPDATE, and ALTER TABLE?**

| Command     | Purpose                                     |
|-------------|---------------------------------------------|
| INSERT      | Adds new records to a table.                |
| SELECT      | Retrieves data from a table.                |
| UPDATE      | Modifies existing records.                  |
| ALTER TABLE | Changes the structure of an existing table. |

**Q7. Why is the WHERE clause important in an UPDATE statement? What could happen if we write an UPDATE statement without a WHERE condition?**

The WHERE clause specifies which records should be updated. If an UPDATE statement is written without a WHERE condition, **all records in the table may be updated.**

**Example:**

```
UPDATE Customers
```

```
SET Remark = 'Regular Buyer'
```

```
WHERE CustomerID = 89790;
```

Here, only the customer with CustomerID = 89790 is updated.